

52

[64]

SEAT No. _____

No. of Printed Pages: 02

SARDAR PATEL UNIVERSITY

F.Y.B.Sc. (CA&IT) || SEMESTER-II- (Regular) Examination-2019

PS02CIIT21 || Advanced C & Introduction to Data Structures

Date: 28/03/2019, Thursday

Time: 10:00am to 01.00pm

Marks: 70

Q. 1 Multiple Choice Questions

[10]

- 1 Which of the following defines a pointer variable to an integer?
[A] int &ptr; [C] int **ptr;
[B] int *ptr; [D] int &&ptr;
- 2 Which of the following is valid? If p1 and p2 are properly declared and initialized and point to same array.
[A] p1/p2 [C] p2 - p1
[B] p1 * p2 [D] p1 / 3
- 3 malloc() function sets the initial value in memory as
[A] zero [C] garbage
[B] NULL [D] None of these
- 4 Which of the following statement is not true about structure?
[A] A field in a structure can itself be a structure. [C] Structure definition is just a template which is used to declare variables of type struct.
[B] Individual members of a structure can be passed to the function. [D] A structure can not have two fields with same type.
- 5 Which of the following allows a portion of memory to be shared by different types of data?
[A] Array [C] Union
[B] Structure [D] File
- 6 Which of the following data structure store the homogeneous data elements??
[A] Arrays [C] Pointers
[B] Records [D] None of these
- 7 When new data are to be inserted into a data structure, but there is no available space; this situation is usually called?
[A] underflow [C] housefull
[B] overflow [D] saturated
- 8 A data structure in which insertion of an element occurs at one end and deletion of an element occurs at other end is known as _____.
[A] Tree [C] Stack
[B] Graph [D] Queues
- 9 A linked list is _____ type of data structure?
[A] Linear [C] Both (A) and (B)
[B] Non-Linear [D] None of the Above
- 10 A linked list in which last node pointing to the first node is known as _____.
[A] Singly linked list [C] Circular linked list
[B] Doubly linked list [D] None of the above

(1)

(P.T.O.)

Q. 2 Short Question (Any TEN)

[20]

- 1 Give the concept of pointers to array.
- 2 Differentiate between '*' and '&' operators in pointers.
- 3 Differentiate malloc() and calloc().
- 4 Define: structure, member operator.
- 5 Explain typedef in brief with suitable example.
- 6 List file modes available to manage the file in C.
- 7 Define and explain in brief: Primitive Data Structure.
- 8 Draw the Hierarchical Structure of Data Structure.
- 9 List operations on a Stack.
- 10 Define: Dqueue.
- 11 Describes the different types of Linked List.
- 12 State various Applications of Linked List.

Q. 3

[A] Define pointer variable. How can we declare and initialize pointer variable? How can we access value of variable through pointer type variable? **[5]**

[B] Write a note on Dynamic memory allocation. **[5]**

OR

[A] Explain pointer arithmetic with example. **[5]**

[B] Write note on: pointer to pointer. **[5]**

Q. 4

[A] Explain array of structures using suitable examples. **[5]**

[B] Explain fprintf and fscanf function with example. **[5]**

OR

[A] What is union? Explain its definition, declaration and assigning values to members of union. **[5]**

[B] Explain the getw and putw function with example. **[5]**

Q. 5

[A] Write a short note on non linear data structure. **[5]**

[B] Write an algorithm to Push & Pop an element into a Stack. **[5]**

OR

[A] Explain the data structure with c briefly. **[5]**

[B] Write an algorithm for Peep & Change operation of a Stack. **[5]**

Q. 6 Write an algorithm to insert & delete an element into a simple queue. **[10]**

OR

Q. 6 Write an algorithm to delete an element from a Singly linked list. **[10]**